

Supplementary Material

Fine-Tuning Diffusion Models for personalized Content Creation

Anonymous submission

A Category-Level Heuristics for Synthetic Data

The synthetic CTR simulation described in Section 3 of the main paper uses per-subcategory constants for the click-to-purchase ratio (CtPR), the base click-through rate, and the joint gender/age distribution. Official statistics for these quantities are not publicly available, so we elicit plausible values with GPT-4o by prompting it with each category–subcategory pair and asking for realistic estimates.

A.1 Click-to-Purchase Ratio (CtPR)

Category	Subcategory	CtPR
accessories	Bags & Luggage	0.08
accessories	Fashion & Silver Jewellery	0.10
accessories	Gold & Diamond Jewellery	0.10
accessories	Handbags & Clutches	0.09
accessories	Jewellery	0.10
accessories	Sunglasses	0.10
accessories	Watches	0.10
appliances	Air Conditioners	0.05
appliances	All Appliances	0.05
appliances	Heating & Cooling Appliances	0.05
appliances	Kitchen & Home Appliances	0.08
appliances	Refrigerators	0.06
appliances	Washing Machines	0.05
bags & luggage	Backpacks	0.09
bags & luggage	Rucksacks	0.08
bags & luggage	Suitcases & Trolley Bags	0.08
bags & luggage	Travel Accessories	0.08
bags & luggage	Travel Duffles	0.08
bags & luggage	Wallets	0.09
beauty & health	Beauty & Grooming	0.11
beauty & health	Diet & Nutrition	0.11
beauty & health	Health & Personal Care	0.10
beauty & health	Household Supplies	0.12
beauty & health	Luxury Beauty	0.11
beauty & health	Make-up	0.12
beauty & health	Personal Care Appliances	0.09
beauty & health	Value Bazaar	0.12
car & motorbike	All Car & Motorbike Products	0.06
car & motorbike	Car & Bike Care	0.07
car & motorbike	Car Accessories	0.07
car & motorbike	Car Electronics	0.07
car & motorbike	Car Parts	0.07
car & motorbike	Motorbike Accessories & Parts	0.06
grocery & gourmet foods	All Grocery & Gourmet Foods	0.12

Table Continued

Category	Subcategory	CtPR
grocery & gourmet foods	Coffee, Tea & Beverages	0.13
grocery & gourmet foods	Snack Foods	0.12
home & kitchen	All Home & Kitchen	0.10
home & kitchen	Bedroom Linen	0.10
home & kitchen	Furniture	0.08
home & kitchen	Garden & Outdoors	0.09
home & kitchen	Home Décor	0.10
home & kitchen	Home Furnishing	0.10
home & kitchen	Home Improvement	0.09
home & kitchen	Home Storage	0.09
home & kitchen	Indoor Lighting	0.09
home & kitchen	Kitchen & Dining	0.10
home & kitchen	Kitchen Storage & Containers	0.09
home & kitchen	Sewing & Craft Supplies	0.09
home, kitchen, pets	Refurbished & Open Box	0.05
industrial supplies	Industrial & Scientific Supplies	0.05
industrial supplies	Janitorial & Sanitation Supplies	0.05
industrial supplies	Lab & Scientific	0.05
industrial supplies	Test, Measure & Inspect	0.05
kids' fashion	Baby Fashion	0.11
kids' fashion	Kids' Clothing	0.11
kids' fashion	Kids' Fashion	0.11
kids' fashion	Kids' Shoes	0.11
kids' fashion	Kids' Watches	0.11
kids' fashion	School Bags	0.11
men's clothing	Innerwear	0.10
men's clothing	Jeans	0.10
men's clothing	Shirts	0.10
men's clothing	T-shirts & Polos	0.10
men's shoes	Casual Shoes	0.10
men's shoes	Formal Shoes	0.10
men's shoes	Sports Shoes	0.10
music	Musical Instruments & Professional Audio	0.06
pet supplies	All Pet Supplies	0.09
pet supplies	Dog supplies	0.09
sports & fitness	All Exercise & Fitness	0.08
sports & fitness	All Sports, Fitness & Outdoors	0.08
sports & fitness	Badminton	0.07
sports & fitness	Camping & Hiking	0.08
sports & fitness	Cardio Equipment	0.08
sports & fitness	Cricket	0.08
sports & fitness	Cycling	0.07
sports & fitness	Fitness Accessories	0.08
sports & fitness	Football	0.08
sports & fitness	Running	0.08
sports & fitness	Strength Training	0.08
sports & fitness	Yoga	0.08
stores	Amazon Fashion	0.10
stores	Fashion Sales & Deals	0.12
stores	Men's Fashion	0.11
stores	Sportswear	0.10
stores	The Designer Boutique	0.12
stores	Women's Fashion	0.11
toys & baby products	Baby Bath, Skin & Grooming	0.12
toys & baby products	Baby Products	0.11

Table Continued

Category	Subcategory	CtPR
toys & baby products	Diapers	0.12
toys & baby products	International Toy Store	0.12
toys & baby products	Nursing & Feeding	0.11
toys & baby products	STEM Toys Store	0.11
toys & baby products	Strollers & Prams	0.12
toys & baby products	Toys & Games	0.11
toys & baby products	Toys Gifting Store	0.11
tv, audio & cameras	All Electronics	0.07
tv, audio & cameras	Camera Accessories	0.07
tv, audio & cameras	Cameras	0.06
tv, audio & cameras	Headphones	0.07
tv, audio & cameras	Home Audio & Theater	0.06
tv, audio & cameras	Home Entertainment Systems	0.06
tv, audio & cameras	Security Cameras	0.07
tv, audio & cameras	Speakers	0.06
tv, audio & cameras	Televisions	0.06
women's clothing	Clothing	0.11
women's clothing	Ethnic Wear	0.11
women's clothing	Lingerie & Nightwear	0.10
women's clothing	Western Wear	0.11
women's shoes	Ballerinas	0.10
women's shoes	Fashion Sandals	0.10
women's shoes	Shoes	0.10

A.2 Base Click-Through Rate

Category	Subcategory	Base CTR
accessories	Bags & Luggage	0.005
accessories	Fashion & Silver Jewellery	0.007
accessories	Gold & Diamond Jewellery	0.007
accessories	Handbags & Clutches	0.006
accessories	Jewellery	0.007
accessories	Sunglasses	0.007
accessories	Watches	0.007
appliances	Air Conditioners	0.0035
appliances	All Appliances	0.004
appliances	Heating & Cooling Appliances	0.003
appliances	Kitchen & Home Appliances	0.004
appliances	Refrigerators	0.0035
appliances	Washing Machines	0.0035
bags & luggage	Backpacks	0.005
bags & luggage	Rucksacks	0.005
bags & luggage	Suitcases & Trolley Bags	0.005
bags & luggage	Travel Accessories	0.005
bags & luggage	Travel Duffles	0.005
bags & luggage	Wallets	0.005
beauty & health	Beauty & Grooming	0.0055
beauty & health	Diet & Nutrition	0.005
beauty & health	Health & Personal Care	0.005
beauty & health	Household Supplies	0.005
beauty & health	Luxury Beauty	0.0055
beauty & health	Make-up	0.0055

Table Continued

Category	Subcategory	Base CTR
beauty & health	Personal Care Appliances	0.0055
beauty & health	Value Bazaar	0.005
car & motorbike	All Car & Motorbike Products	0.003
car & motorbike	Car & Bike Care	0.003
car & motorbike	Car Accessories	0.003
car & motorbike	Car Electronics	0.003
car & motorbike	Car Parts	0.003
car & motorbike	Motorbike Accessories & Parts	0.003
grocery & gourmet foods	All Grocery & Gourmet Foods	0.006
grocery & gourmet foods	Coffee, Tea & Beverages	0.0055
grocery & gourmet foods	Snack Foods	0.0055
home & kitchen	All Home & Kitchen	0.0045
home & kitchen	Bedroom Linen	0.004
home & kitchen	Furniture	0.004
home & kitchen	Garden & Outdoors	0.004
home & kitchen	Home Décor	0.004
home & kitchen	Home Furnishing	0.004
home & kitchen	Home Improvement	0.004
home & kitchen	Home Storage	0.004
home & kitchen	Indoor Lighting	0.004
home & kitchen	Kitchen & Dining	0.004
home & kitchen	Kitchen Storage & Containers	0.004
home & kitchen	Sewing & Craft Supplies	0.004
home, kitchen, pets	Refurbished & Open Box	0.003
industrial supplies	Industrial & Scientific Supplies	0.003
industrial supplies	Janitorial & Sanitation Supplies	0.003
industrial supplies	Lab & Scientific	0.003
industrial supplies	Test, Measure & Inspect	0.003
kids' fashion	Baby Fashion	0.0075
kids' fashion	Kids' Clothing	0.007
kids' fashion	Kids' Fashion	0.007
kids' fashion	Kids' Shoes	0.007
kids' fashion	Kids' Watches	0.007
kids' fashion	School Bags	0.007
men's clothing	Innerwear	0.0065
men's clothing	Jeans	0.0065
men's clothing	Shirts	0.0065
men's clothing	T-shirts & Polos	0.0065
men's shoes	Casual Shoes	0.0065
men's shoes	Formal Shoes	0.0065
men's shoes	Sports Shoes	0.0065
music	Musical Instruments & Professional Audio	0.003
pet supplies	All Pet Supplies	0.0045
pet supplies	Dog supplies	0.0045
sports & fitness	All Exercise & Fitness	0.005
sports & fitness	All Sports, Fitness & Outdoors	0.005
sports & fitness	Badminton	0.0045
sports & fitness	Camping & Hiking	0.005
sports & fitness	Cardio Equipment	0.0045
sports & fitness	Cricket	0.0045
sports & fitness	Cycling	0.0045
sports & fitness	Fitness Accessories	0.005
sports & fitness	Football	0.0045
sports & fitness	Running	0.0045
sports & fitness	Strength Training	0.005

Table Continued

Category	Subcategory	Base CTR
sports & fitness	Yoga	0.005
stores	Amazon Fashion	0.0065
stores	Fashion Sales & Deals	0.007
stores	Men's Fashion	0.0065
stores	Sportswear	0.0065
stores	The Designer Boutique	0.007
stores	Women's Fashion	0.0065
toys & baby products	Baby Bath, Skin & Grooming	0.007
toys & baby products	Baby Products	0.0075
toys & baby products	Diapers	0.0075
toys & baby products	International Toy Store	0.007
toys & baby products	Nursing & Feeding	0.0075
toys & baby products	STEM Toys Store	0.007
toys & baby products	Strollers & Prams	0.0075
toys & baby products	Toys & Games	0.007
toys & baby products	Toys Gifting Store	0.007
tv, audio & cameras	All Electronics	0.004
tv, audio & cameras	Camera Accessories	0.0035
tv, audio & cameras	Cameras	0.0035
tv, audio & cameras	Headphones	0.0035
tv, audio & cameras	Home Audio & Theater	0.0035
tv, audio & cameras	Home Entertainment Systems	0.0035
tv, audio & cameras	Security Cameras	0.0035
tv, audio & cameras	Speakers	0.0035
tv, audio & cameras	Televisions	0.0035
women's clothing	Clothing	0.006
women's clothing	Ethnic Wear	0.0065
women's clothing	Lingerie & Nightwear	0.0065
women's clothing	Western Wear	0.0065
women's shoes	Ballerinas	0.006
women's shoes	Fashion Sandals	0.0065
women's shoes	Shoes	0.0065

A.3 Joint Gender and Age Distribution

Category / Subcategory	Male	0–17	18–24	25–34	35–44	45–54	55–64	65+
accessories								
Bags & Luggage	0.50	0.15	0.30	0.30	0.15	0.07	0.02	0.01
Fashion & Silver Jewellery	0.30	0.10	0.30	0.30	0.20	0.07	0.02	0.01
Gold & Diamond Jewellery	0.30	0.05	0.20	0.30	0.25	0.15	0.03	0.02
Handbags & Clutches	0.20	0.08	0.30	0.35	0.20	0.05	0.02	0.01
Jewellery	0.40	0.05	0.20	0.30	0.25	0.15	0.03	0.02
Sunglasses	0.40	0.15	0.35	0.30	0.15	0.05	0.02	0.01
Watches	0.40	0.15	0.25	0.25	0.15	0.15	0.07	0.01
appliances								
Air Conditioners	0.60	0.10	0.25	0.25	0.20	0.10	0.07	0.03
All Appliances	0.65	0.08	0.22	0.25	0.25	0.12	0.06	0.02
Heating & Cooling Appliances	0.50	0.08	0.22	0.30	0.25	0.10	0.03	0.02
Kitchen & Home Appliances	0.34	0.12	0.30	0.28	0.18	0.08	0.03	0.01
Refrigerators	0.45	0.12	0.30	0.28	0.20	0.08	0.02	0.01
Washing Machines	0.50	0.12	0.30	0.30	0.20	0.08	0.02	0.01
bags & luggage								

Table Continued

Category / Subcategory	Male	0–17	18–24	25–34	35–44	45–54	55–64	65+
Backpacks	0.55	0.20	0.35	0.30	0.10	0.03	0.01	0.01
Rucksacks	0.50	0.10	0.28	0.30	0.20	0.08	0.03	0.01
Suitcases & Trolley Bags	0.38	0.12	0.30	0.35	0.15	0.05	0.02	0.01
Travel Accessories	0.45	0.12	0.30	0.35	0.15	0.05	0.02	0.01
Travel Duffles	0.48	0.12	0.30	0.35	0.15	0.05	0.02	0.01
Wallets	0.35	0.12	0.30	0.30	0.15	0.10	0.02	0.01
beauty & health								
Beauty & Grooming	0.30	0.12	0.35	0.30	0.15	0.05	0.02	0.01
Diet & Nutrition	0.50	0.12	0.30	0.30	0.15	0.08	0.03	0.02
Health & Personal Care	0.25	0.10	0.25	0.30	0.20	0.10	0.03	0.02
Household Supplies	0.22	0.10	0.30	0.30	0.20	0.07	0.02	0.01
Luxury Beauty	0.21	0.05	0.25	0.35	0.25	0.08	0.02	0.01
Make-up	0.12	0.05	0.25	0.40	0.20	0.07	0.02	0.01
Personal Care Appliances	0.19	0.12	0.30	0.30	0.20	0.05	0.02	0.01
Value Bazaar	0.30	0.08	0.25	0.35	0.25	0.05	0.02	0.01
car & motorbike								
All Car & Motorbike Products	0.70	0.15	0.30	0.25	0.15	0.10	0.03	0.02
Car & Bike Care	0.75	0.20	0.30	0.25	0.15	0.05	0.03	0.02
Car Accessories	0.70	0.15	0.30	0.30	0.15	0.07	0.02	0.01
Car Electronics	0.85	0.15	0.32	0.30	0.15	0.05	0.02	0.01
Car Parts	0.88	0.18	0.30	0.28	0.15	0.05	0.03	0.01
Motorbike Accessories & Parts	0.70	0.20	0.40	0.25	0.10	0.03	0.01	0.01
grocery & gourmet foods								
All Grocery & Gourmet Foods	0.40	0.10	0.25	0.30	0.20	0.10	0.03	0.02
Coffee, Tea & Beverages	0.40	0.15	0.30	0.25	0.20	0.07	0.02	0.01
Snack Foods	0.42	0.08	0.25	0.30	0.25	0.10	0.02	0.01
home & kitchen								
All Home & Kitchen	0.50	0.12	0.25	0.28	0.20	0.10	0.03	0.02
Bedroom Linen	0.40	0.10	0.25	0.30	0.25	0.07	0.02	0.01
Furniture	0.28	0.10	0.25	0.30	0.20	0.10	0.03	0.02
Garden & Outdoors	0.39	0.12	0.28	0.30	0.18	0.07	0.03	0.02
Home Décor	0.20	0.10	0.25	0.30	0.20	0.10	0.03	0.02
Home Furnishing	0.30	0.08	0.22	0.30	0.25	0.10	0.03	0.02
Home Improvement	0.29	0.12	0.25	0.28	0.20	0.10	0.03	0.02
Home Storage	0.30	0.10	0.25	0.30	0.20	0.10	0.03	0.02
Indoor Lighting	0.50	0.12	0.28	0.30	0.18	0.07	0.03	0.02
Kitchen & Dining	0.30	0.10	0.28	0.30	0.20	0.08	0.03	0.01
Kitchen Storage & Containers	0.35	0.10	0.25	0.30	0.20	0.10	0.03	0.02
Sewing & Craft Supplies	0.30	0.10	0.25	0.30	0.25	0.08	0.02	0.01
home, kitchen, pets								
Refurbished & Open Box	0.25	0.12	0.30	0.30	0.18	0.07	0.02	0.01
industrial supplies								
Industrial & Scientific Supplies	0.50	0.15	0.30	0.30	0.15	0.07	0.02	0.01
Janitorial & Sanitation Supplies	0.50	0.15	0.30	0.30	0.15	0.07	0.02	0.01
Lab & Scientific	0.50	0.15	0.30	0.30	0.15	0.07	0.02	0.01
Test, Measure & Inspect	0.50	0.15	0.30	0.30	0.15	0.07	0.02	0.01
kids' fashion								
Baby Fashion	0.35	0.10	0.30	0.35	0.15	0.07	0.02	0.01
Kids' Clothing	0.28	0.08	0.30	0.35	0.20	0.05	0.02	0.01
Kids' Fashion	0.24	0.08	0.30	0.35	0.20	0.05	0.02	0.01
Kids' Shoes	0.25	0.08	0.30	0.35	0.20	0.05	0.02	0.01
Kids' Watches	0.26	0.08	0.30	0.35	0.20	0.05	0.02	0.01
School Bags	0.31	0.10	0.30	0.35	0.20	0.05	0.02	0.01
men's clothing								
Innerwear	0.70	0.12	0.35	0.30	0.15	0.05	0.02	0.01

Table Continued

Category / Subcategory	Male	0–17	18–24	25–34	35–44	45–54	55–64	65+
Jeans	0.80	0.12	0.40	0.30	0.15	0.05	0.02	0.01
Shirts	0.80	0.15	0.40	0.30	0.10	0.03	0.01	0.01
T-shirts & Polos	0.80	0.15	0.40	0.30	0.10	0.03	0.01	0.01
men's shoes								
Casual Shoes	0.90	0.18	0.40	0.30	0.08	0.02	0.01	0.01
Formal Shoes	0.95	0.15	0.35	0.30	0.15	0.03	0.01	0.01
Sports Shoes	0.80	0.15	0.35	0.30	0.15	0.05	0.02	0.01
music								
Musical Instruments & Professional Audio	0.55	0.10	0.30	0.35	0.15	0.07	0.02	0.01
pet supplies								
All Pet Supplies	0.45	0.08	0.22	0.30	0.25	0.10	0.03	0.02
Dog Supplies	0.50	0.08	0.25	0.30	0.25	0.07	0.03	0.02
sports & fitness								
All Exercise & Fitness	0.60	0.18	0.30	0.25	0.15	0.07	0.03	0.02
All Sports, Fitness & Outdoors	0.65	0.20	0.35	0.25	0.12	0.05	0.02	0.01
Badminton	0.50	0.15	0.40	0.25	0.10	0.05	0.03	0.02
Camping & Hiking	0.75	0.18	0.35	0.30	0.12	0.03	0.01	0.01
Cardio Equipment	0.50	0.20	0.35	0.25	0.12	0.05	0.02	0.01
Cricket	0.70	0.25	0.35	0.25	0.10	0.03	0.01	0.01
Cycling	0.65	0.20	0.35	0.25	0.10	0.07	0.02	0.01
Fitness Accessories	0.60	0.20	0.35	0.25	0.10	0.05	0.03	0.02
Football	0.80	0.25	0.40	0.25	0.05	0.03	0.01	0.01
Running	0.55	0.12	0.35	0.30	0.15	0.05	0.02	0.01
Strength Training	0.65	0.17	0.30	0.30	0.15	0.05	0.02	0.01
Yoga	0.30	0.12	0.30	0.35	0.15	0.05	0.02	0.01
stores								
Amazon Fashion	0.40	0.18	0.35	0.25	0.12	0.05	0.03	0.02
Fashion Sales & Deals	0.40	0.15	0.35	0.30	0.15	0.03	0.01	0.01
Men's Fashion	0.81	0.15	0.35	0.30	0.15	0.05	0.02	0.01
Sportswear	0.53	0.12	0.30	0.35	0.15	0.05	0.02	0.01
The Designer Boutique	0.32	0.05	0.20	0.35	0.25	0.10	0.03	0.02
Women's Fashion	0.11	0.05	0.25	0.30	0.25	0.13	0.02	0.01
toys & baby products								
Baby Bath, Skin & Grooming	0.30	0.05	0.20	0.30	0.25	0.15	0.03	0.02
Baby Products	0.40	0.05	0.18	0.35	0.30	0.10	0.02	0.01
Diapers	0.35	0.05	0.20	0.35	0.25	0.10	0.03	0.02
International Toy Store	0.32	0.08	0.30	0.35	0.20	0.05	0.02	0.01
Nursing & Feeding	0.21	0.08	0.25	0.35	0.25	0.05	0.02	0.01
STEM Toys Store	0.41	0.08	0.30	0.35	0.20	0.05	0.02	0.01
Strollers & Prams	0.32	0.08	0.25	0.35	0.25	0.05	0.02	0.01
Toys & Games	0.31	0.08	0.30	0.35	0.20	0.05	0.02	0.01
Toys Gifting Store	0.35	0.08	0.30	0.35	0.20	0.05	0.02	0.01
tv, audio & cameras								
All Electronics	0.65	0.12	0.28	0.30	0.18	0.07	0.03	0.02
Camera Accessories	0.70	0.15	0.35	0.30	0.12	0.05	0.02	0.01
Cameras	0.75	0.12	0.30	0.35	0.15	0.05	0.02	0.01
Headphones	0.50	0.25	0.30	0.25	0.12	0.05	0.02	0.01
Home Audio & Theater	0.55	0.12	0.28	0.30	0.18	0.07	0.03	0.02
Home Entertainment Systems	0.50	0.12	0.30	0.30	0.18	0.07	0.02	0.01
Security Cameras	0.60	0.15	0.35	0.30	0.15	0.05	0.02	0.01
Speakers	0.56	0.12	0.35	0.30	0.15	0.05	0.02	0.01
Televisions	0.56	0.12	0.35	0.30	0.15	0.05	0.02	0.01
women's clothing								
Clothing	0.15	0.10	0.40	0.30	0.10	0.05	0.03	0.02

Table Continued

Category / Subcategory	Male	0–17	18–24	25–34	35–44	45–54	55–64	65+
Ethnic Wear	0.20	0.08	0.35	0.30	0.15	0.07	0.03	0.02
Lingerie & Nightwear	0.10	0.05	0.20	0.35	0.25	0.10	0.03	0.02
Western Wear	0.10	0.15	0.25	0.25	0.25	0.08	0.02	0.01
women’s shoes								
Ballerinas	0.05	0.15	0.40	0.25	0.10	0.05	0.03	0.02
Fashion Sandals	0.05	0.10	0.40	0.30	0.15	0.03	0.01	0.01
Shoes	0.10	0.08	0.30	0.35	0.20	0.05	0.02	0.01

B Reward Model Hyperparameter Sweep

We select DeepFM hyperparameters via a sequential grid search following the protocol of the original paper. All models are trained with Adam, ReLU activations, and a 90%/10% train/validation split on the reward-training half of the dataset.

Batch size

Batch size	Log loss	AUC
1024	0.0543	0.6922
2048	0.0543	0.6926
4096	0.0545	0.6927

Learning rate

Learning rate	Log loss	AUC
5×10^{-4}	0.0543	0.6926
1×10^{-3}	0.0543	0.6926
3×10^{-3}	0.0548	0.6802

Dropout rate

Dropout	Log loss	AUC
0.0	0.0543	0.6908
0.1	0.0543	0.6921
0.2	0.0543	0.6924
0.3	0.0543	0.6920
0.4	0.0543	0.6924
0.5	0.0543	0.6926
0.6	0.0544	0.6913
0.7	0.0545	0.6906
0.8	0.0547	0.6880
0.9	0.0546	0.6894
1.0 (FM only)	0.0545	0.6888

Hidden-layer width

Neurons per layer	Log loss	AUC
100	0.0543	0.6919
200	0.0544	0.6907
400	0.0543	0.6926
800	0.0545	0.6938

Hidden-layer depth

# Layers	Log loss	AUC
1	0.0543	0.6929
3	0.0545	0.6938
5	0.0544	0.6902
7	0.0547	0.6895

Layer shape (3-layer configurations)

Shape	Log loss	AUC
Constant (800, 800, 800)	0.0545	0.6938
Increasing (400, 800, 1200)	0.0543	0.6926
Decreasing (1200, 800, 400)	0.0543	0.6922
Diamond (600, 1200, 600)	0.0546	0.6921

FM latent dimension

Latent dim.	Log loss	AUC
4	0.0543	0.6921
8	0.0543	0.6934
10	0.0543	0.6926
16	0.0546	0.6869
32	0.0546	0.6874
64	0.0546	0.6870

C Demographic Regularization Grid Search

We sweep regularization strengths λ_{gender} and λ_{age} first independently (with the other set to zero), then jointly over the most promising values.

Gender regularization ($\lambda_{\text{age}} = 0$)

λ_{gender}	Log loss	AUC
0.01	0.0545	0.6883
0.05	0.0548	0.6862
0.1	0.0547	0.6852
0.5	0.0546	0.6893
1.0	0.0545	0.6873
2.0	0.0545	0.6879

Age regularization ($\lambda_{\text{gender}} = 0$)

λ_{age}	Log loss	AUC
0.01	0.0546	0.6859
0.05	0.0547	0.6873
0.1	0.0545	0.6880
0.5	0.0546	0.6863
1.0	0.0547	0.6893
2.0	0.0545	0.6877

Joint regularization

$(\lambda_{\text{gender}}, \lambda_{\text{age}})$	Log loss	AUC
(0.5, 0.5)	0.0550	0.6891
(1.0, 0.5)	0.0545	0.6884
(0.5, 1.0)	0.0545	0.6890
(1.0, 1.0)	0.0545	0.6899

D Diffusion Fine-Tuning Configuration

DDPO-IS hyperparameters

Component	Hyperparameter	Value
Diffusion	Denoising steps (T)	50
	Classifier-free guidance weight	5.0
Optimization	Optimizer	AdamW
	Learning rate	1×10^{-5}
	Weight decay	1×10^{-4}
	β_1, β_2	0.9, 0.999
	ϵ	1×10^{-8}
	Gradient clip norm	1.0
DDPO-IS	Batch size	64
	Samples per iteration	256
	Gradient updates per iteration	4
	IS clip range	1×10^{-4}